YOUNG INVENTORS WHO CHANGED THE

20 True Stories of Kids Like You (Ages 8-14)

20 Inventor Stories * Science * Your Turn Challenges * Worksheets

By Daniel Tesfamariam

YOUNG INVENTORS WHO CHANGED THE WORLD

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INVENTOR #1

Louis Braille

Age: 15 | France, 1824

Invention: Reading System for the Blind

THE STORY:

Louis lost his sight in an accident at age 3. At school for the blind, he found the existing raised-letter system impossible to use. Inspired by a military night-writing code, Louis spent 2 years (ages 13-15) developing a system of 6 raised dots that could represent every letter, number, and punctuation mark. Today, Braille is used worldwide!

IMPACT:

Over 39 million blind people worldwide

THE SCIENCE:

`Tactile communication, pattern design, combinatorics`

YOUR TURN: INVENTOR #1

INVENTION CHALLENGE:

Design a new way to help people with a disability.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #2

Frank Epperson

Age: 11 | USA, 1905

Invention: The Popsicle (Ice Pop)

THE STORY:

On a cold San Francisco night, 11-year-old Frank accidentally left a cup of soda mix with a stirring stick outside. By morning, it had frozen solid around the stick! He called it an ’Epsicle.’ Years later, he patented it as the Popsicle. Today, over 2 billion popsicles are sold yearly!

IMPACT:

Happy accident turned into billion-dollar industry

THE SCIENCE:

`Chemistry of freezing, accidental discovery, observation`

YOUR TURN: INVENTOR #2

INVENTION CHALLENGE:

What everyday item could you improve with a simple change?

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #3

Philo Farnsworth

Age: 14 | USA, 1920

Invention: Electronic Television

THE STORY:

At 14, while plowing a potato field in rows, Philo had a brilliant idea: what if you could transmit images in rows of light, line by line? He sketched his concept for his high school teacher. By 21, he demonstrated the first fully electronic television. His Idaho potato field inspiration changed entertainment forever!

IMPACT:

Television exists in 1.7 billion households worldwide

THE SCIENCE:

`Electron beams, cathode rays, image scanning`

YOUR TURN: INVENTOR #3

INVENTION CHALLENGE:

Invent something using only items from your recycling bin.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #4

Gitanjali Rao

Age: 11 | USA, 2017

Invention: Lead Water Contamination Detector

THE STORY:

After watching news about the Flint, Michigan water crisis, 11-year-old Gitanjali was determined to help. She invented a device called Tethys that detects lead in water faster and cheaper than existing methods. Using carbon nanotube sensors connected to a smartphone app, her invention can test water in seconds!

IMPACT:

37 million Americans are served by water systems with lead violations

THE SCIENCE:

`Nanotechnology, sensor design, app development`

YOUR TURN: INVENTOR #4

INVENTION CHALLENGE:

Design a device that detects something harmful in the environment.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #5

Jack Andraka

Age: 15 | USA, 2012

Invention: Pancreatic Cancer Test

THE STORY:

After losing a family friend to pancreatic cancer, 15-year-old Jack researched why the disease is so deadly (usually detected too late). He invented a paper sensor that detects pancreatic cancer 168 times faster, 26,000 times cheaper, and 400 times more sensitive than existing tests. He was rejected by 197 labs before one said yes!

IMPACT:

Could save thousands of lives through early detection

THE SCIENCE:

`Bioengineering, carbon nanotubes, protein markers`

YOUR TURN: INVENTOR #5

INVENTION CHALLENGE:

Create a faster or cheaper way to test for a health problem.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #6

Kelvin Doe

Age: 13 | Sierra Leone, 2012

Invention: Generator & Radio Station from Trash

THE STORY:

In a country with severe electricity problems, 13-year-old Kelvin built a generator from discarded electronics found in trash bins. He then built his own radio station to broadcast music and news in his community! He became the youngest person invited to visit MIT’s engineering program.

IMPACT:

Provided electricity and media to his entire community

THE SCIENCE:

`Electrical engineering, resourcefulness, circuit design`

YOUR TURN: INVENTOR #6

INVENTION CHALLENGE:

Build something useful from items others have thrown away.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #7

Param Jaggi

Age: 14 | USA, 2011

Invention: Device to Convert CO_2 to Oxygen

THE STORY:

Concerned about car emissions, 14-year-old Param invented a device that attaches to car tailpipes and uses algae to convert carbon dioxide into oxygen through photosynthesis. His Algae Mobile device showed that young people can tackle even the biggest environmental challenges.

IMPACT:

Cars produce 4.6 metric tons of CO_2 per year each

THE SCIENCE:

`Photosynthesis, algae biology, emission reduction`

YOUR TURN: INVENTOR #7

INVENTION CHALLENGE:

Design an invention that helps the environment using nature.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #8

Elif Bilgin

Age: 16 | Turkey, 2013

Invention: Bioplastic from Banana Peels

THE STORY:

Elif spent 2 years experimenting to create a bioplastic from banana peels as an alternative to petroleum-based plastics. After many failed attempts, she succeeded in creating a material that could replace some traditional plastics. Her persistence through failure is as inspiring as her invention!

IMPACT:

World uses 380 million tons of plastic per year

THE SCIENCE:

`Polymer chemistry, sustainable materials, perseverance`

YOUR TURN: INVENTOR #8

INVENTION CHALLENGE:

Create a useful material from food waste.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #9

Kenneth Shinozuka

Age: 15 | USA, 2014

Invention: Wearable Sensor for Alzheimer's Patients

THE STORY:

After his grandfather with Alzheimer's wandered from home, Kenneth invented a wearable sensor that alerts caregivers when patients get out of bed. The pressure sensor in a sock sends Bluetooth alerts to a smartphone. Simple, affordable, and life-saving for millions of families dealing with Alzheimer's.

IMPACT:

Over 6 million Americans have Alzheimer's disease

THE SCIENCE:

`Sensors, Bluetooth, app development, user-centered design`

YOUR TURN: INVENTOR #9

INVENTION CHALLENGE:

Invent a device that helps elderly or sick people stay safe.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #10

Hannah Herbst

Age: 15 | USA, 2015

Invention: Ocean Energy Harvesting Device

THE STORY:

Hannah’s pen pal in Ethiopia lived without electricity. Inspired to help, Hannah invented BEACON: a device that harvests energy from ocean currents to power small devices and provide light. Using a 3D-printed propeller and generator, her prototype proved the concept of accessible ocean energy.

IMPACT:

940 million people worldwide lack access to electricity

THE SCIENCE:

`Renewable energy, fluid dynamics, 3D printing`

YOUR TURN: INVENTOR #10

INVENTION CHALLENGE:

Design something that harvests energy from an unusual source.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #11

Mikaila Ulmer

Age: 9 | USA, 2009

Invention: Me & the Bees Lemonade

THE STORY:

After being stung by a bee twice in one week, 9-year-old Mikaila didn’t get angry - she got curious! She learned about the importance of bees and their decline. She started a lemonade company using her grandmother’s recipe with honey, donating a percentage to save bees. She got a $60K deal on Shark Tank at age 11!

IMPACT:

Bees pollinate 75% of flowering plants and 35% of food crops

THE SCIENCE:

`Entrepreneurship, environmental conservation, business`

YOUR TURN: INVENTOR #11

INVENTION CHALLENGE:

Start a kid business that solves a problem AND helps the world.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #12

Cassidy Goldstein

Age: 12 | USA, 2003

Invention: Unbreakable Crayon Holder

THE STORY:

Frustrated that her crayons kept breaking (especially the small pieces too tiny to hold), 12-year-old Cassidy invented a plastic tube holder that grips broken crayon pieces, allowing them to be used until completely finished. She patented her invention and it sold in stores!

IMPACT:

Reduces waste - no more thrown-away broken crayons

THE SCIENCE:

`Product design, patents, problem observation`

YOUR TURN: INVENTOR #12

INVENTION CHALLENGE:

Improve an everyday item that frustrates you.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #13

Ann Makosinski

Age: 15 | Canada, 2013

Invention: Body-Heat Powered Flashlight

THE STORY:

When Ann learned her friend in the Philippines couldn’t study at night due to no electricity, she invented a flashlight powered only by body heat! Using Peltier tiles that convert temperature difference into electricity, her ’Hollow Flashlight’ needs no batteries or charging. Just hold it!

IMPACT:

Design for people without access to electricity or batteries

THE SCIENCE:

`Thermoelectric effect, energy conversion, humanitarian design`

YOUR TURN: INVENTOR #13

INVENTION CHALLENGE:

Invent something powered by a surprising energy source.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #14

Ryan Patterson

Age: 17 | USA, 2001

Invention: Sign Language Translator Glove

THE STORY:

Ryan invented a glove embedded with sensors that translates American Sign Language finger-spelling into text displayed on a screen. Each finger movement is detected and matched to a letter, enabling deaf individuals to communicate more easily with hearing people who don’t know sign language.

IMPACT:

Over 500,000 people use ASL in North America

THE SCIENCE:

`Sensors, pattern recognition, assistive technology`

YOUR TURN: INVENTOR #14

INVENTION CHALLENGE:

Design a device that helps people communicate better.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #15

Moziah Bridges

Age: 9 | USA, 2011

Invention: Mo’s Bows - Handmade Bow Ties

THE STORY:

When 9-year-old Mo couldn’t find bow ties he liked, he taught himself to sew using YouTube and his grandmother’s guidance. He started selling handmade bow ties and built a million-dollar business by age 15! He appeared on Shark Tank and was mentored by Daymond John.

IMPACT:

Youth entrepreneurship proves age is not a barrier

THE SCIENCE:

`Fashion design, entrepreneurship, self-teaching`

YOUR TURN: INVENTOR #15

INVENTION CHALLENGE:

Create a product from a skill you taught yourself.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #16

Keiana Cave

Age: 16 | USA, 2014

Invention: Chemical Test for Environmental Pollution

THE STORY:

Keiana developed a method to detect cancer-causing compounds in water from industrial contamination. Her research helped identify pollution in her New Orleans community that was making people sick. She proved that young scientists can contribute to environmental justice.

IMPACT:

Environmental pollution causes 12.6 million deaths per year globally

THE SCIENCE:

`Analytical chemistry, environmental science, activism`

YOUR TURN: INVENTOR #16

INVENTION CHALLENGE:

Invent something that detects pollution or harmful substances.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #17

Taylor Wilson

Age: 14 | USA, 2008

Invention: Nuclear Fusion Reactor

THE STORY:

Taylor became the youngest person ever to build a working nuclear fusion reactor at age 14! His device smashes atoms together, demonstrating the same process that powers the Sun. He went on to develop nuclear detection technology for national security.

IMPACT:

Youngest nuclear scientist in history

THE SCIENCE:

`Nuclear physics, fusion, plasma physics, radiation`

YOUR TURN: INVENTOR #17

INVENTION CHALLENGE:

Design an experiment that demonstrates a physics concept.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #18

Tripp Phillips

Age: 8 | USA, 2014

Invention: Le-Glue (Temporary Lego Adhesive)

THE STORY:

Tired of his LEGO creations falling apart, 8-year-old Tripp invented Le-Glue: a non-toxic, water-soluble adhesive that holds LEGO bricks together but dissolves with water when you want to take them apart. His product has sold over a million units!

IMPACT:

Solves a problem every LEGO builder has experienced

THE SCIENCE:

`Chemistry, product development, market identification`

YOUR TURN: INVENTOR #18

INVENTION CHALLENGE:

Solve a problem that every kid you know experiences.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

INVENTOR #19

Kiara Nirghin

Age: 16 | South Africa, 2016

Invention: Super Absorbent Material for Drought

THE STORY:

During South Africa's worst drought in decades, Kiara invented a low-cost super absorbent polymer made from orange peels and avocado skins. Placed in soil, it retains hundreds of times its weight in water, reducing irrigation needs by up to 73% and helping farmers survive drought.

IMPACT:

Agriculture uses 70% of global freshwater; droughts are increasing

THE SCIENCE:

`Polymer science, agriculture, waste utilization`

YOUR TURN: INVENTOR #19

INVENTION CHALLENGE:

Invent something that uses food/plant waste productively.

MY INVENTION IDEA:

Problem I want to solve: ______________________________

My solution: ______________________________

Materials needed: ______________________________

How it works: ______________________________

Who would use it: ______________________________

SKETCH YOUR INVENTION:

INVENTOR #20

Bishop Curry V

Age: 10 | USA, 2016

Invention: Oasis - Device to Prevent Hot Car Deaths

THE STORY:

After hearing about a baby who died in a hot car, 10-year-old Bishop invented Oasis: a device that monitors temperature inside cars and sends alerts to parents' phones. It can also activate cooling fans when temperatures rise dangerously. His empathy-driven innovation could save lives.

IMPACT:

Average of 38 children die from hot cars per year in the US

THE SCIENCE:

`Temperature sensors, IoT, alerts, safety engineering`

YOUR TURN: INVENTOR #20

INVENTION CHALLENGE:

Design a safety device that protects children.

MY INVENTION IDEA:

Problem I want to solve: ____

My solution: ____

Materials needed: ____

How it works: ____

Who would use it: ____

SKETCH YOUR INVENTION:

THE SCIENTIFIC METHOD

1. OBSERVE - Notice a problem or ask a question

2. RESEARCH - Learn what’s already been tried

3. HYPOTHESIZE - Predict a possible solution

4. EXPERIMENT - Test your idea

5. ANALYZE - Study your results

6. CONCLUDE - Did it work? What did you learn?

7. SHARE - Tell others about your discovery!

MY INVENTION WORKSHEET

Inventor name: ____

Problem to solve: ____

Who has this problem: ____

My solution idea: ____

Materials needed: ____

How it works (explain): ____

How to test it: ____

Expected result: ____

YOUNG INVENTOR CERTIFICATE

Awarded to:

____

For completing this entire book!

Date: ________________

Author: Daniel Tesfamariam

MY INVENTORS JOURNAL - Entry 1

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 2

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 3

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 4

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 5

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 6

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 7

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 8

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 9

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY INVENTORS JOURNAL - Entry 10

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

INVENTORS ACTIVITY #1

CHALLENGE:

Think of 3 new ideas related to this topic and explain each one.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #2

CHALLENGE:

Create a mind map connecting everything you learned.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #3

CHALLENGE:

Write a letter explaining this topic to a younger child.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #4

CHALLENGE:

Design a poster that teaches the most important concept.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #5

CHALLENGE:

List 10 questions you still have about this topic.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #6

CHALLENGE:

Draw a comic strip showing what you learned in action.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

INVENTORS ACTIVITY #7

CHALLENGE:

Write 5 ways you can apply this knowledge in real life.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

CREATIVE SPACE: INVENTORS

Draw what you learned today

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: INVENTORS

Illustrate your favorite concept

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: INVENTORS

Create a visual summary of this chapter

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: INVENTORS

Draw yourself using this knowledge

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: INVENTORS

Design a symbol that represents this topic

Use this space for drawing, writing, or brainstorming!

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS

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